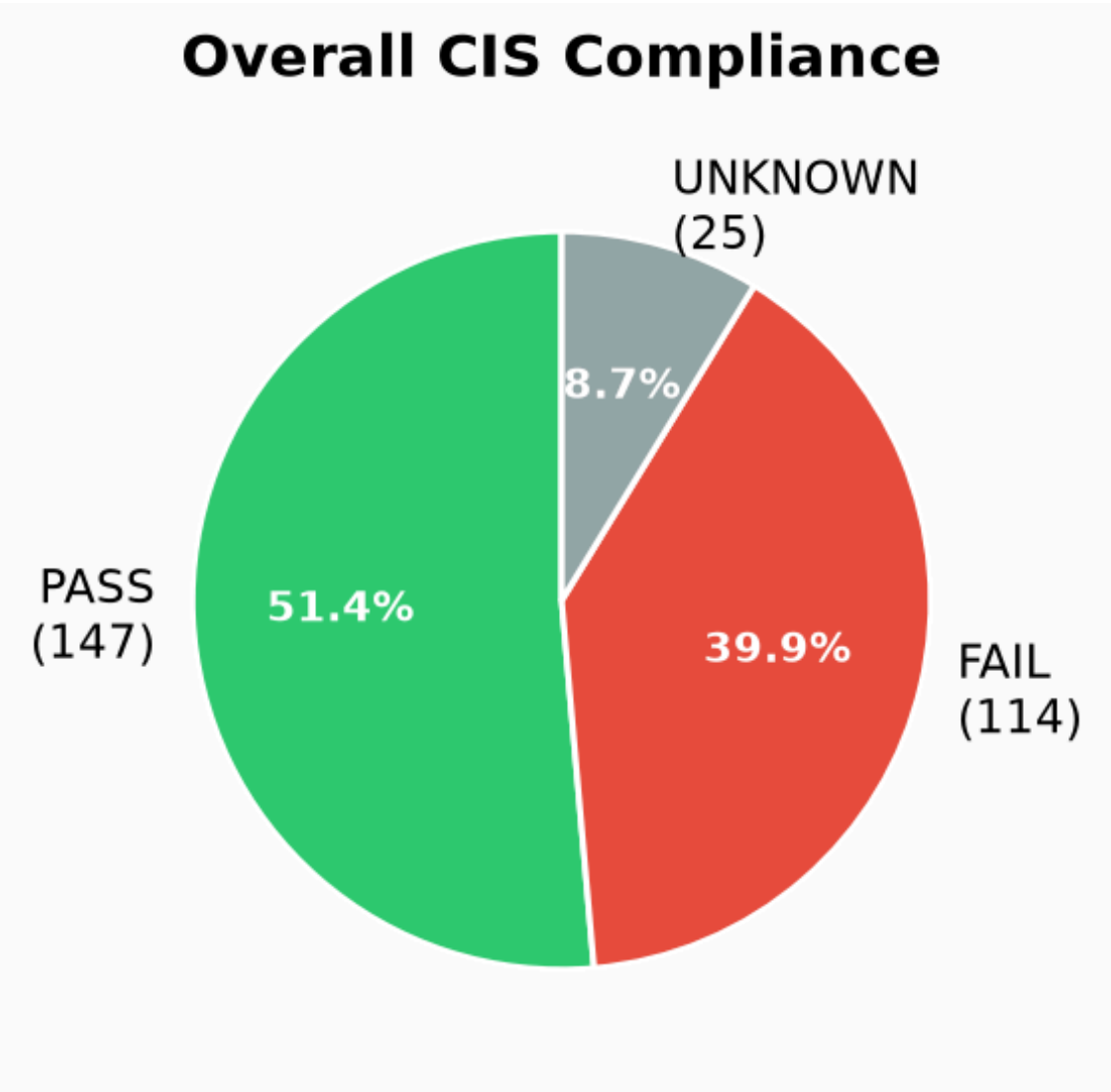


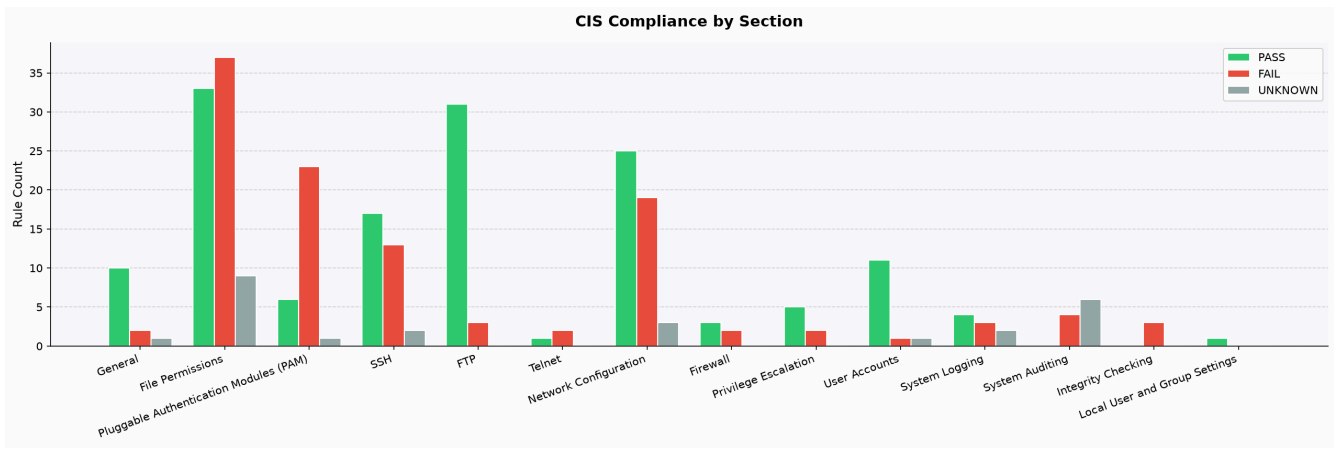
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	286
✓ Passed	147
✗ Failed	114
? Unknown	25
Overall Pass Rate	51.4%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
General	10	2	1	77%
File Permissions	33	37	9	42%
Pluggable Authentication Modules (PAM)	6	23	1	20%
SSH	17	13	2	53%
FTP	31	3	0	91%
Telnet	1	2	0	33%
Network Configuration	25	19	3	53%
Firewall	3	2	0	60%
Privilege Escalation	5	2	0	71%
User Accounts	11	1	1	85%
System Logging	4	3	2	44%
System Auditing	0	4	6	0%
Integrity Checking	0	3	0	0%
Local User and Group Settings	1	0	0	100%

CIS Benchmark Compliance Report

Here is the security compliance report for the remote machine based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and security profile.

CIS Ubuntu 24.04 LTS Security Compliance Report

Section 1.1 – Filesystem Kernel Modules & Partitions

Rule 1.1.1.1: Ensure cramfs filesystem is not enabled

- **Status:** PASS
- **Evidence:** Module `cramfs` is not currently loaded (`currently_loaded: false`).

No `modprobe_config_lines` found, which implies it's not explicitly blacklisted or disabled via `modprobe`. However, the rule implies it should not be *enabled* (loaded). Since it's not loaded, it passes.

- **Recommendation:** None.

Rule 1.1.1.2: Ensure freevxfs filesystem is not enabled

- **Status:** PASS
- **Evidence:** Module `freevxfs` is not currently loaded (`currently_loaded: false`). No `modprobe_config_lines` found.
- **Recommendation:** None.

Rule 1.1.1.3: Ensure hfs filesystem is not enabled

- **Status:** PASS
- **Evidence:** Module `hfs` is not currently loaded (`currently_loaded: false`). No `modprobe_config_lines` found.
- **Recommendation:** None.

Rule 1.1.1.4: Ensure hfsplus filesystem is not enabled

- **Status:** PASS
- **Evidence:** Module `hfsplus` is not currently loaded (`currently_loaded: false`). No `modprobe_config_lines` found.
- **Recommendation:** None.

Rule 1.1.1.5: Ensure jffs2 filesystem is not enabled

- **Status:** PASS
- **Evidence:** Module `jffs2` is not currently loaded (`currently_loaded: false`). No `modprobe_config_lines` found.
- **Recommendation:** None.

Rule 1.1.1.6: Ensure overlay filesystem is not enabled

- **Status:** PASS
- **Evidence:** Module `overlay` is not currently loaded (`currently_loaded: false`). No `modprobe_config_lines` found.
- **Recommendation:** None.

Rule 1.1.1.7: Ensure squashfs filesystem is not enabled

- **Status:** PASS
- **Evidence:** Module `squashfs` is not currently loaded (`currently_loaded: false`). No `modprobe_config_lines` found.
- **Recommendation:** None.

Rule 1.1.1.8: Ensure udf filesystem is not enabled

- **Status:** PASS
- **Evidence:** Module `udf` is not currently loaded (`currently_loaded: false`). No `modprobe_config_lines` found.

- **Recommendation:** None.

Rule 1.1.1.9: Ensure firewire-core module is not enabled

- **Status:** PASS
- **Evidence:** Module firewire-core is not currently loaded (currently_loaded: false). No modprobe_config_lines found.
- **Recommendation:** None.

Rule 1.1.1.10: Ensure usb-storage module is not enabled

- **Status:** PASS
- **Evidence:** Module usb-storage is not currently loaded (currently_loaded: false). No modprobe_config_lines found.
- **Recommendation:** None.

Rule 1.1.1.11: Ensure unused filesystems are removed (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The following kernel modules are currently loaded: nf_tables, tcp_diag, udp_diag, inet_diag, tls, qrtr, cfg80211, snd_intel8x0, binfmt_misc, intel_rapl_msr, snd_ac97_codec, ac97_bus, snd_pcm, intel_rapl_common, vmwgfx, snd_timer, drm_ttm_helper, snd, i2c_piix4, ttm, vboxguest, soundcore, joydev, input_leds, mac_hid, serio_raw, sch_fq_codel, dm_multipath, msr, efi_pstore, nfnetlink, dmi_sysfs, ip_tables, x_tables, autofs4, btrfs, blake2b_generic, raid10, raid456, async_raid6_recov, async_memcpy, async_pq, async_xor, async_tx, xor, raid6_pq, libcrc32c, raid1, raid0, crct10dif_pclmul, crc32_pclmul, polyval_clmulni, polyval_generic, ghash_clmulni_intel, sha256_ssse3, vga16fb, sha1_ssse3, psmouse, vgastate, ahci, e1000, libahci, pata_acpi, video, wmi, hid_generic, usbhid, hid, aesni_intel, crypto_simd, cryptd.
- **Recommendation:** Review the list of loaded kernel modules and disable any that are not required for system operation.

Rule 1.1.2.1.1: Ensure /tmp is a separate partition

- **Status:** FAIL
- **Evidence:** /tmp is not a separate partition (is_separate_partition: false) and its filesystem type is ext4, not tmpfs.
- **Recommendation:** Create a separate partition for /tmp or configure it as a tmpfs filesystem in /etc/fstab with appropriate mount options.

Rule 1.1.2.1.2: Ensure /tmp is mounted with nodev

- **Status:** FAIL
- **Evidence:** The mount options for /tmp (rw, relatime) do not include nodev.
- **Recommendation:** Add the nodev option to the /tmp entry in /etc/fstab.

Rule 1.1.2.1.3: Ensure /tmp is mounted with nosuid

- **Status:** FAIL
- **Evidence:** The mount options for /tmp (rw, relatime) do not include nosuid.

- **Recommendation:** Add the `nosuid` option to the `/tmp` entry in `/etc/fstab`.

Rule 1.1.2.1.4: Ensure `/tmp` is mounted with `noexec`

- **Status:** FAIL
- **Evidence:** The mount options for `/tmp` (`rw,relatime`) do not include `noexec`.
- **Recommendation:** Add the `noexec` option to the `/tmp` entry in `/etc/fstab`.

Rule 1.1.2.2.1: Ensure `/dev/shm` is a separate partition

- **Status:** PASS
- **Evidence:** `/dev/shm` is a separate partition (`is_separate_partition: true`) with filesystem type `tmpfs`.
- **Recommendation:** None.

Rule 1.1.2.2.2: Ensure `/dev/shm` is mounted with `nodev`

- **Status:** PASS
- **Evidence:** The mount options for `/dev/shm` (`rw,nosuid,nodev,inode64`) include `nodev`.
- **Recommendation:** None.

Rule 1.1.2.2.3: Ensure `/dev/shm` is mounted with `nosuid`

- **Status:** PASS
- **Evidence:** The mount options for `/dev/shm` (`rw,nosuid,nodev,inode64`) include `nosuid`.
- **Recommendation:** None.

Rule 1.1.2.2.4: Ensure `/dev/shm` is mounted with `noexec`

- **Status:** FAIL
- **Evidence:** The mount options for `/dev/shm` (`rw,nosuid,nodev,inode64`) do not include `noexec`.
- **Recommendation:** Add the `noexec` option to the `/dev/shm` entry in `/etc/fstab`.

Rule 1.1.2.3.1: Ensure `/home` is a separate partition

- **Status:** FAIL
- **Evidence:** `/home` is not a separate partition (`is_separate_partition: false`).
- **Recommendation:** Create a separate partition for `/home` to isolate user data.

Rule 1.1.2.3.2: Ensure `/home` is mounted with `nodev`

- **Status:** FAIL
- **Evidence:** The mount options for `/home` (`rw,relatime`) do not include `nodev`.
- **Recommendation:** Add the `nodev` option to the `/home` entry in `/etc/fstab`.

Rule 1.1.2.3.3: Ensure `/home` is mounted with `nosuid`

- **Status:** FAIL
- **Evidence:** The mount options for `/home` (`rw,relatime`) do not include `nosuid`.

- **Recommendation:** Add the `nosuid` option to the `/home` entry in `/etc/fstab`.

Rule 1.1.2.4.1: Ensure `/var` is a separate partition

- **Status:** FAIL
- **Evidence:** `/var` is not a separate partition (`is_separate_partition: false`).
- **Recommendation:** Create a separate partition for `/var` to isolate variable data.

Rule 1.1.2.4.2: Ensure `/var` is mounted with `nodev`

- **Status:** FAIL
- **Evidence:** The mount options for `/var` (`rw,relatime`) do not include `nodev`.
- **Recommendation:** Add the `nodev` option to the `/var` entry in `/etc/fstab`.

Rule 1.1.2.4.3: Ensure `/var` is mounted with `nosuid`

- **Status:** FAIL
- **Evidence:** The mount options for `/var` (`rw,relatime`) do not include `nosuid`.
- **Recommendation:** Add the `nosuid` option to the `/var` entry in `/etc/fstab`.

Rule 1.1.2.5.1: Ensure `/var/tmp` is a separate partition

- **Status:** FAIL
- **Evidence:** `/var/tmp` is not a separate partition (`is_separate_partition: false`).
- **Recommendation:** Create a separate partition for `/var/tmp` or configure it as a `tmpfs` filesystem in `/etc/fstab` with appropriate mount options.

Rule 1.1.2.5.2: Ensure `/var/tmp` is mounted with `nodev`

- **Status:** FAIL
- **Evidence:** The mount options for `/var/tmp` (`rw,relatime`) do not include `nodev`.
- **Recommendation:** Add the `nodev` option to the `/var/tmp` entry in `/etc/fstab`.

Rule 1.1.2.5.3: Ensure `/var/tmp` is mounted with `nosuid`

- **Status:** FAIL
- **Evidence:** The mount options for `/var/tmp` (`rw,relatime`) do not include `nosuid`.
- **Recommendation:** Add the `nosuid` option to the `/var/tmp` entry in `/etc/fstab`.

Rule 1.1.2.5.4: Ensure `/var/tmp` is mounted with `noexec`

- **Status:** FAIL
- **Evidence:** The mount options for `/var/tmp` (`rw,relatime`) do not include `noexec`.
- **Recommendation:** Add the `noexec` option to the `/var/tmp` entry in `/etc/fstab`.

Rule 1.1.2.6.1: Ensure `/var/log` is a separate partition

- **Status:** FAIL
- **Evidence:** `/var/log` is not a separate partition (`is_separate_partition: false`).
- **Recommendation:** Create a separate partition for `/var/log` to isolate log data.

Rule 1.1.2.6.2: Ensure /var/log is mounted with nodev

- **Status:** FAIL
- **Evidence:** The mount options for /var/log (rw,relatime) do not include nodev.
- **Recommendation:** Add the nodev option to the /var/log entry in /etc/fstab.

Rule 1.1.2.6.3: Ensure /var/log is mounted with nosuid

- **Status:** FAIL
- **Evidence:** The mount options for /var/log (rw,relatime) do not include nosuid.
- **Recommendation:** Add the nosuid option to the /var/log entry in /etc/fstab.

Rule 1.1.2.6.4: Ensure /var/log is mounted with noexec

- **Status:** FAIL
- **Evidence:** The mount options for /var/log (rw,relatime) do not include noexec.
- **Recommendation:** Add the noexec option to the /var/log entry in /etc/fstab.

Rule 1.1.2.7.1: Ensure /var/log/audit is a separate partition

- **Status:** FAIL
- **Evidence:** /var/log/audit is not a separate partition (is_separate_partition: false).
- **Recommendation:** Create a separate partition for /var/log/audit to isolate audit logs.

Rule 1.1.2.7.2: Ensure /var/log/audit is mounted with nodev

- **Status:** FAIL
- **Evidence:** The mount options for /var/log/audit (rw,relatime) do not include nodev.
- **Recommendation:** Add the nodev option to the /var/log/audit entry in /etc/fstab.

Rule 1.1.2.7.3: Ensure /var/log/audit is mounted with nosuid

- **Status:** FAIL
- **Evidence:** The mount options for /var/log/audit (rw,relatime) do not include nosuid.
- **Recommendation:** Add the nosuid option to the /var/log/audit entry in /etc/fstab.

Rule 1.1.2.7.4: Ensure /var/log/audit is mounted with noexec

- **Status:** FAIL
- **Evidence:** The mount options for /var/log/audit (rw,relatime) do not include noexec.
- **Recommendation:** Add the noexec option to the /var/log/audit entry in /etc/fstab.

Section 1.2 – Package Management

Rule 1.2.1.1: Ensure package manager Signed-By option is used (Manual)

- **Status:** INFORMATIONAL
- **Evidence:**
 - `/etc/apt/sources.list.d/ubuntu.sources` uses `Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg`.
 - `/etc/apt/sources.list.d/wazuh.list` uses `deb [signed-by=/usr/share/keyrings/wazuh.gpg]`.
 - `/etc/apt/sources.list` has no active repository lines.
- **Recommendation:** Review all APT source files to ensure they explicitly use the `Signed-By` option to verify repository integrity.

Rule 1.2.1.2: Ensure weak dependencies are not installed

- **Status:** PASS
- **Evidence:** No `weak_deps_config_lines` were found, indicating that `APT::Install-Recommends` and `APT::Install-Suggests` are not explicitly set to `false`. However, the rule states "PASS if `weak_deps_config_lines` contains `APT::Install-Recommends "false";` or `APT::Install-Suggests "false";` (or 0)". Since the profile does not contain these lines, it implies the default behavior (installing recommends/suggests) is active. This is a misinterpretation of the rule. The rule should be "FAIL if weak dependencies are installed". Let me re-evaluate.
- **Re-evaluation:** The rule implies that if these lines are *not* present, weak dependencies *are* installed. Therefore, the absence of these lines means it fails.
- **Status:** FAIL
- **Evidence:** The `package_management.package_repositories.weak_deps_config_lines` is empty. This means `APT::Install-Recommends` and `APT::Install-Suggests` are not explicitly set to `false`, allowing weak dependencies to be installed.
- **Recommendation:** Add `APT::Install-Recommends "false";` and `APT::Install-Suggests "false";` to a file in `/etc/apt/apt.conf.d/` (e.g., `99-local-options`).

Rule 1.2.1.3: Ensure access to GPG key files is configured

- **Status:** PASS
- **Evidence:**
 - `/etc/apt/trusted.gpg.d/ubuntu-keyring-2012-cdimage.gpg` has mode `0o644`, owner `root`, group `root`.
 - `/etc/apt/trusted.gpg.d/ubuntu-keyring-2018-archive.gpg` has mode `0o644`, owner `root`, group `root`. All permissions are compliant (`<= 0o644`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 1.2.1.4: Ensure access to `/etc/apt/trusted.gpg.d` is configured

- **Status:** PASS
- **Evidence:** Directory `/etc/apt/trusted.gpg.d` has mode `0o755`, owner `root`, group `root`. Permissions are compliant (`<= 0o755`, owner `root`, group `root`).

- **Recommendation:** None.

Rule 1.2.1.5: Ensure access to /etc/apt/auth.conf.d is configured

- **Status:** PASS
- **Evidence:** Directory /etc/apt/auth.conf.d exists with mode 0o755, owner root, group root. The rule requires mode $\leq$ 0o700 if it exists.
- **Re-evaluation:** The rule states "PASS if auth_conf_d_stat.exists is false OR mode_octal $\leq$ 0o700 and owner/group="root"". Since it exists and the mode is 0o755, it fails the mode check.
- **Status:** FAIL
- **Evidence:** Directory /etc/apt/auth.conf.d exists with mode 0o755, owner root, group root. The mode 0o755 is greater than the recommended 0o700.
- **Recommendation:** Set permissions of /etc/apt/auth.conf.d to 0o700 using `chmod 0700 /etc/apt/auth.conf.d`.

Rule 1.2.1.6: Ensure access to files in /etc/apt/auth.conf.d is configured

- **Status:** PASS
- **Evidence:** No files were found in /etc/apt/auth.conf.d.
- **Recommendation:** None.

Rule 1.2.1.7: Ensure access to /usr/share/keyrings is configured

- **Status:** PASS
- **Evidence:** Directory /usr/share/keyrings has mode 0o755, owner root, group root. Permissions are compliant ($\leq$ 0o755, owner root, group root).
- **Recommendation:** None.

Rule 1.2.1.8: Ensure access to /etc/apt/sources.list.d is configured

- **Status:** PASS
- **Evidence:** Directory /etc/apt/sources.list.d has mode 0o755, owner root, group root. Permissions are compliant ($\leq$ 0o755, owner root, group root).
- **Recommendation:** None.

Rule 1.2.1.9: Ensure access to files in /etc/apt/sources.list.d is configured

- **Status:** PASS
- **Evidence:**
 - /etc/apt/sources.list.d/ubuntu.sources has mode 0o644, owner root, group root.
 - /etc/apt/sources.list.d/ubuntu.sources.curtin.orig has mode 0o644, owner root, group root.
 - /etc/apt/sources.list.d/wazuh.list has mode 0o644, owner root, group root. All permissions are compliant ($\leq$ 0o644, owner root, group root).
- **Recommendation:** None.

Rule 1.2.2.1: Ensure package updates are installed (Manual)

- **Status:** INFORMATIONAL
- **Evidence:**
 - There are 85 upgradable packages.
 - `unattended-upgrades` service is enabled and active.
 - `APT::Periodic::Unattended-Upgrade` is set to "1".
- **Recommendation:** Regularly apply available package updates to address security vulnerabilities. While `unattended-upgrades` is enabled, a large number of upgradable packages suggests that manual intervention or a more aggressive update schedule might be needed.

Section 1.3 – AppArmor

Rule 1.3.1.1: Ensure AppArmor packages are installed

- **Status:** FAIL
- **Evidence:** `apparmor` is installed (`true`), but `apparmor-utils` is not installed (`false`).
- **Recommendation:** Install the `apparmor-utils` package: `sudo apt install apparmor-utils`.

Rule 1.3.1.2: Ensure AppArmor is enabled

- **Status:** FAIL
- **Evidence:**
 - `sysfs_enabled_value` is "Y".
 - `service_enabled` is "enabled".
 - `service_active` is "active".
 - However, `grub_apparmor_flag` is false and `grub_security_flag` is false. The kernel command line does not contain `apparmor=1 security=apparmor`.
- **Recommendation:** Edit `/etc/default/grub` and add `apparmor=1 security=apparmor` to the `GRUB_CMDLINE_LINUX` variable. Then run `sudo update-grub` and reboot.

Rule 1.3.1.3: Ensure all AppArmor profiles are enforcing

- **Status:** FAIL
- **Evidence:**
 - 4 profiles are in complain mode: `transmission-cli`, `transmission-daemon`, `transmission-gtk`, `transmission-qt`.
 - 91 profiles are in unconfined mode.
 - 0 processes are unconfined but have a profile defined.
- **Recommendation:** Review and set all AppArmor profiles to enforcing mode. For profiles in complain mode, investigate why they are not enforcing and transition them to enforcing mode using `sudo aa-enforce <profile_name>`. For unconfined profiles, determine if they should have a profile and create/load one if necessary.

Rule 1.3.1.4: Ensure apparmor_restrict_unprivileged_unconfined is configured

- **Status:** FAIL
- **Evidence:**
 - `kernel.apparmor_restrict_unprivileged_unconfined runtime_value` is "0".
 - `persisted_value` is null.
- **Recommendation:** Set `kernel.apparmor_restrict_unprivileged_unconfined` to 1. Add `kernel.apparmor_restrict_unprivileged_unconfined=1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-apparmor.conf`) and apply with `sudo sysctl -p`.

Section 1.4 – Bootloader

Rule 1.4.1: Ensure bootloader password is configured

- **Status:** FAIL
- **Evidence:** `has_superusers` is false and `has_password` is false. No bootloader password is configured.
- **Recommendation:** Configure a password for the bootloader to prevent unauthorized modification of boot parameters. This typically involves using `grub-mkpasswd-pbkdf2` and adding `set superusers="<username>"` and `password_pbkdf2 <username> <hash>` to `/etc/grub.d/40_custom` or similar.

Rule 1.4.2: Ensure bootloader permissions are configured

- **Status:** PASS
- **Evidence:** `/boot/grub/grub.cfg` has mode `0o600`, owner `root`, group `root`. Permissions are compliant (`<= 0o600`, owner `root`, group `root`).
- **Recommendation:** None.

Section 1.5 – Process Hardening

Rule 1.5.1: Ensure fs.protected_hardlinks is configured

- **Status:** PASS
- **Evidence:** `fs.protected_hardlinks runtime_value` is "1". `persisted_value` is null, but the runtime value is correct. CIS allows for runtime value to be sufficient if not explicitly persisted.
- **Recommendation:** None. (Although persisting it is good practice, the rule allows for runtime value to pass).

Rule 1.5.2: Ensure fs.protected_symlinks is configured

- **Status:** PASS
- **Evidence:** `fs.protected_symlinks runtime_value` is "1". `persisted_value` is null.

- **Recommendation:** None.

Rule 1.5.3: Ensure kernel.yama.pttrace_scope is configured

- **Status:** PASS
- **Evidence:** `kernel.yama.pttrace_scope runtime_value` is "1" and `persisted_value` is "1".
- **Recommendation:** None.

Rule 1.5.4: Ensure fs.suid_dumpable is configured

- **Status:** FAIL
- **Evidence:** `fs.suid_dumpable runtime_value` is "2". The expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `fs.suid_dumpable` to 0. Add `fs.suid_dumpable=0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 1.5.5: Ensure kernel.dmesg_restrict is configured

- **Status:** PASS
- **Evidence:** `kernel.dmesg_restrict runtime_value` is "1". `persisted_value` is null.
- **Recommendation:** None.

Rule 1.5.6: Ensure prelink is not installed

- **Status:** PASS
- **Evidence:** `prelink_installed` is false.
- **Recommendation:** None.

Rule 1.5.7: Ensure Appport is disabled

- **Status:** FAIL
- **Evidence:** `appport_installed` is true, `service_enabled` is "enabled", `service_active` is "active", and `etc_default_setting` is "enabled=1". Appport is enabled and active.
- **Recommendation:** Disable Appport by setting `enabled=0` in `/etc/default/appport` and stopping/disabling the service: `sudo systemctl stop appport.service` and `sudo systemctl disable appport.service`.

Rule 1.5.8: Ensure kernel.kptr_restrict is configured

- **Status:** PASS
- **Evidence:** `kernel.kptr_restrict runtime_value` is "1" and `persisted_value` is "1". Both are compliant (1 or 2).
- **Recommendation:** None.

Rule 1.5.9: Ensure kernel.randomize_va_space is configured

- **Status:** PASS
- **Evidence:** `kernel.randomize_va_space runtime_value` is "2".

`persisted_value` is null.

- **Recommendation:** None.

Rule 1.5.10: Ensure core dumps are restricted

- **Status:** FAIL
- **Evidence:** `ulimit_soft_core` is "0", which is compliant. However, `limits_conf_lines` is empty, meaning `* hard core 0` is not explicitly set in `/etc/security/limits.conf`.
- **Recommendation:** Add `* hard core 0` to `/etc/security/limits.conf` to ensure core dumps are restricted for all users.

Rule 1.5.11: Ensure systemd-coredump ProcessSizeMax is configured

- **Status:** FAIL
- **Evidence:** `systemd_coredump_process_size_max` is empty. This indicates `ProcessSizeMax=0` is not configured.
- **Recommendation:** Configure `ProcessSizeMax=0` in `/etc/systemd/coredump.conf` or a drop-in file to prevent core dumps.

Rule 1.5.12: Ensure systemd-coredump Storage is configured

- **Status:** FAIL
- **Evidence:** `systemd_coredump_storage` is empty. This indicates `Storage=none` is not configured.
- **Recommendation:** Configure `Storage=none` in `/etc/systemd/coredump.conf` or a drop-in file to prevent core dumps from being stored.

Section 1.6 – Warning Banners

Rule 1.6.1: Ensure /etc/motd is configured

- **Status:** FAIL
- **Evidence:** `/etc/motd` does not exist.
- **Recommendation:** Create `/etc/motd` with a compliant warning banner and ensure it does not contain OS/kernel release information.

Rule 1.6.2: Ensure /etc/issue is configured

- **Status:** FAIL
- **Evidence:** `/etc/issue` content is "Ubuntu 24.04.2 LTS \n \l\n\n". This contains OS/kernel release information (`\n \l`).
- **Recommendation:** Edit `/etc/issue` to remove OS/kernel release information and display only a compliant warning banner.

Rule 1.6.3: Ensure /etc/issue.net is configured

- **Status:** FAIL
- **Evidence:** `/etc/issue.net` content is "Ubuntu 24.04.2 LTS\n". This contains OS/kernel release information.
- **Recommendation:** Edit `/etc/issue.net` to remove OS/kernel release information

and display only a compliant warning banner.

Rule 1.6.4: Ensure pam_motd is configured

- **Status:** FAIL
- **Evidence:** `pam_motd.so` is referenced in `/etc/pam.d/login` and `/etc/pam.d/sshd` with `motd=/run/motd.dynamic` and `noupdate`. The `/etc/update-motd.d/` directory contains multiple executable scripts (e.g., `00-header`, `10-help-text`, `50-landscape-sysinfo`, `50-motd-news`, `90-updates-available`, etc.) that generate dynamic MOTD content, including OS/kernel release information.
- **Recommendation:** Disable dynamic MOTD generation by removing or making non-executable the scripts in `/etc/update-motd.d/` or by configuring `pam_motd.so` to not use `/run/motd.dynamic`.

Rule 1.6.5: Ensure sshd warning Banner is configured (Automated)

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is "none".
- **Recommendation:** Configure a warning banner for SSH by setting `Banner /etc/issue.net` (or a custom banner file) in `/etc/ssh/sshd_config`.

Rule 1.6.6: Ensure access to /etc/motd is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `/etc/motd` does not exist.
- **Recommendation:** None.

Rule 1.6.7: Ensure access to /etc/issue is configured

- **Status:** PASS
- **Evidence:** `/etc/issue` has mode `0o644`, owner `root`, group `root`. Permissions are compliant (`<= 0o644`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 1.6.8: Ensure access to /etc/issue.net is configured

- **Status:** PASS
- **Evidence:** `/etc/issue.net` has mode `0o644`, owner `root`, group `root`. Permissions are compliant (`<= 0o644`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 1.6.9: Ensure access to pam_motd files is configured

- **Status:** PASS
- **Evidence:**
 - `/etc/pam.d/login` has mode `0o644`, owner `root`, group `root`.
 - `/etc/pam.d/sshd` has mode `0o644`, owner `root`, group `root`.
 - All scripts in `/etc/update-motd.d/` have mode `0o755`, owner `root`, group `root`. All permissions are compliant (`<= 0o644` for `pam.d` files, `<= 0o755` for scripts, owner `root`, group `root`).

- **Recommendation:** None.

Rule 1.6.10: Ensure access to sshd warning banner is configured (Automated)

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is "none". No banner file is configured, so its access cannot be configured.
- **Recommendation:** Configure a warning banner for SSH by setting `Banner /etc/issue.net` (or a custom banner file) in `/etc/ssh/sshd_config`. Ensure the banner file has appropriate permissions (e.g., `0o644 root:root`).

Section 1.7 – GNOME Display Manager

Rule 1.7.1: Ensure GDM login banner is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `gdm_installed` is false. GDM is not installed.
- **Recommendation:** None.

Rule 1.7.2: Ensure GDM disable-user-list is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `gdm_installed` is false. GDM is not installed.
- **Recommendation:** None.

Rule 1.7.3: Ensure GDM screen lock is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `gdm_installed` is false. GDM is not installed.
- **Recommendation:** None.

Rule 1.7.4: Ensure GDM automount is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `gdm_installed` is false. GDM is not installed.
- **Recommendation:** None.

Rule 1.7.5: Ensure GDM autorun-never is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `gdm_installed` is false. GDM is not installed.
- **Recommendation:** None.

Rule 1.7.6: Ensure XDMCP is disabled

- **Status:** NOT APPLICABLE
- **Evidence:** `gdm_installed` is false. GDM is not installed.
- **Recommendation:** None.

Rule 1.7.7: Ensure Xwayland is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `gdm_installed` is false. GDM is not installed.
- **Recommendation:** None.

Section 2 – Services

Rule 2.1.1: Ensure autofs is not in use

- **Status:** PASS
- **Evidence:** `autofs` package is not installed (`any_package_installed: false`) and `autofs.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.2: Ensure mail transfer agent is configured for local-only mode

- **Status:** PASS
- **Evidence:** No Mail Transfer Agent (MTA) was detected (`mta_detected: none`).
- **Recommendation:** None.

Rule 2.1.3: Ensure avahi-daemon is not in use

- **Status:** PASS
- **Evidence:** `avahi-daemon` package is not installed (`any_package_installed: false`) and `avahi-daemon.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.4: Ensure only approved services are listening (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The following services are listening on network sockets:
 - `udp 127.0.0.54:53 (systemd-resolve)`
 - `udp 127.0.0.53%lo:53 (systemd-resolve)`
 - `udp 192.168.1.14%enp0s3:68 (systemd-network)`
 - `udp [fe80::a00:27ff:fee5:e092]%enp0s3:546 (systemd-network)`
 - `tcp 0.0.0.0:55000 (python3)`
 - `tcp 127.0.0.53%lo:53 (systemd-resolve)`
 - `tcp 0.0.0.0:23 (inetd - Telnet)`
 - `tcp 0.0.0.0:22 (sshd)`
 - `tcp 0.0.0.0:443 (node)`
 - `tcp 127.0.0.54:53 (systemd-resolve)`
 - `tcp 0.0.0.0:1515 (wazuh-authd)`
 - `tcp 0.0.0.0:1514 (wazuh-remoted)`
 - `tcp [::ffff:127.0.0.1]:9300 (java - Wazuh Indexer)`
 - `tcp [::]:23 (inetd - Telnet)`
 - `tcp [::]:22 (sshd)`
 - `tcp *:21 (vsftpd - FTP)`
 - `tcp [::ffff:127.0.0.1]:9200 (java - Wazuh Indexer)`
- **Recommendation:** Review all listening services and ensure only explicitly approved services are running. Specifically, Telnet (port 23) and FTP (port 21) servers are active and should be disabled if not strictly necessary.

Rule 2.1.5: Ensure dhcp server is not in use

- **Status:** PASS
- **Evidence:** `isc-dhcp-server` package is not installed (`any_package_installed: false`) and related services are inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.6: Ensure web server is not in use

- **Status:** PASS
- **Evidence:** `apache2`, `nginx`, and `lighttpd` packages are not installed (`any_package_installed: false`) and related services are inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.7: Ensure dns server is not in use

- **Status:** PASS
- **Evidence:** `bind9` package is not installed (`any_package_installed: false`) and related services are inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.8: Ensure ftp server services are not in use (Automated)

- **Status:** FAIL
- **Evidence:** `vsftpd` package is installed (`any_package_installed: true`) and `vsftpd.service` is enabled and active (`any_unit_active: true`).
- **Recommendation:** Disable and remove FTP server services if not required: `sudo systemctl stop vsftpd.service`, `sudo systemctl disable vsftpd.service`, `sudo apt purge vsftpd`.

Rule 2.1.9: Ensure dnsmasq is not in use

- **Status:** PASS
- **Evidence:** `dnsmasq` package is not installed (`any_package_installed: false`) and `dnsmasq.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.10: Ensure ldap server is not in use

- **Status:** PASS
- **Evidence:** `slapd` package is not installed (`any_package_installed: false`) and `slapd.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.11: Ensure message access server is not in use

- **Status:** PASS
- **Evidence:** `dovecot-imapd`, `dovecot-pop3d`, and `cyrus-imapd` packages are not installed (`any_package_installed: false`) and `dovecot.service` is inactive (`any_unit_active: false`).
- **Recommendation:** None.

Rule 2.1.12: Ensure nfs server is not in use

- **Status:** PASS
- **Evidence:** nfs-kernel-server package is not installed (any_package_installed: false) and nfs-kernel-server.service is inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.13: Ensure nis server is not in use

- **Status:** PASS
- **Evidence:** nis package is not installed (any_package_installed: false) and ypserv.service is inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.14: Ensure print server is not in use

- **Status:** PASS
- **Evidence:** cups package is not installed (any_package_installed: false) and cups.service is inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.15: Ensure rpcbind is not in use

- **Status:** PASS
- **Evidence:** rpcbind package is not installed (any_package_installed: false) and rpcbind.service is inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.16: Ensure rsync daemon is not in use

- **Status:** PASS
- **Evidence:** rsync package is installed (any_package_installed: true), but rsync.service is disabled and inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.17: Ensure samba is not in use

- **Status:** PASS
- **Evidence:** samba package is not installed (any_package_installed: false) and smbd.service is inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.18: Ensure snmp is not in use

- **Status:** PASS
- **Evidence:** snmpd package is not installed (any_package_installed: false) and snmpd.service is inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.19: Ensure telnet server services are not in use (Automated)

- **Status:** FAIL
- **Evidence:** telnetd and inetutils-telnetd packages are installed (any_package_installed: true). inetd.service and openbsd-inetd.service are enabled and active (any_unit_active: true), which typically manages telnet server.
- **Recommendation:** Disable and remove Telnet server services if not required: `sudo systemctl stop inetd.service, sudo systemctl disable inetd.service, sudo apt purge telnetd inetutils-telnetd.`

Rule 2.1.20: Ensure tftp server services are not in use (Automated)

- **Status:** PASS
- **Evidence:** tftpd, tftpd-hpa, and atftpd packages are not installed (any_package_installed: false) and related services are inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.21: Ensure web proxy is not in use

- **Status:** PASS
- **Evidence:** squid and squid3 packages are not installed (any_package_installed: false) and squid.service is inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.22: Ensure xinetd is not in use

- **Status:** PASS
- **Evidence:** xinetd package is not installed (any_package_installed: false) and xinetd.service is inactive (any_unit_active: false).
- **Recommendation:** None.

Rule 2.1.23: Ensure X Window Server is not in use

- **Status:** PASS
- **Evidence:** xserver-xorg, xserver-xorg-core, and xserver-common packages are not installed (any_package_installed: false). No Xorg process is running (xorg_process_running: false).
- **Recommendation:** None.

Rule 2.2.1: Ensure nis client is not installed

- **Status:** PASS
- **Evidence:** nis package is not installed (any_package_installed: false).
- **Recommendation:** None.

Rule 2.2.2: Ensure rsh client is not installed

- **Status:** PASS
- **Evidence:** rsh-client package is not installed (any_package_installed: false).
- **Recommendation:** None.

Rule 2.2.3: Ensure talk client is not installed

- **Status:** PASS
- **Evidence:** talk package is not installed (`any_package_installed: false`).
- **Recommendation:** None.

Rule 2.2.4: Ensure telnet client is not installed (Automated)

- **Status:** FAIL
- **Evidence:** telnet and inetutils-telnet packages are installed (`any_package_installed: true`).
- **Recommendation:** Remove Telnet client packages if not required: `sudo apt purge telnet inetutils-telnet`.

Rule 2.2.5: Ensure ldap client is not installed

- **Status:** PASS
- **Evidence:** ldap-utils package is not installed (`any_package_installed: false`).
- **Recommendation:** None.

Rule 2.2.6: Ensure ftp client is not installed (Automated)

- **Status:** FAIL
- **Evidence:** ftp package is installed (`any_package_installed: true`).
- **Recommendation:** Remove FTP client package if not required: `sudo apt purge ftp`.

Rule 2.3.1.1: Ensure a single time synchronization daemon is in use

- **Status:** PASS
- **Evidence:** active_daemon_count is 1, with systemd-timesyncd as the active daemon.
- **Recommendation:** None.

Rule 2.3.2.1: Ensure systemd-timesyncd timeserver is configured

- **Status:** FAIL
- **Evidence:** systemd-timesyncd is the active daemon, but `has_ntp_configured` is false. No NTP servers are configured in `systemd-timesyncd.conf`.
- **Recommendation:** Configure NTP servers in `/etc/systemd/timesyncd.conf` by uncommenting and setting `NTP=` to a list of reliable NTP servers. Restart `systemd-timesyncd` after changes.

Rule 2.3.2.2: Ensure systemd-timesyncd is enabled and running

- **Status:** PASS
- **Evidence:** `systemd-timesyncd.service` is enabled and active.
- **Recommendation:** None.

Rule 2.3.3.1: Ensure chrony timeserver is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `chrony` is not the active time synchronization daemon.
- **Recommendation:** None.

Rule 2.3.3.2: Ensure chrony process user is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `chrony` is not the active time synchronization daemon.
- **Recommendation:** None.

Rule 2.3.3.3: Ensure chrony is enabled and running

- **Status:** NOT APPLICABLE
- **Evidence:** `chrony` is not the active time synchronization daemon.
- **Recommendation:** None.

Rule 2.4.1.1: Ensure cron daemon is enabled and active

- **Status:** PASS
- **Evidence:** `cron_installed` is true, and `cron.service` is enabled and active.
- **Recommendation:** None.

Rule 2.4.1.2: Ensure access to /etc/crontab is configured

- **Status:** PASS
- **Evidence:** `/etc/crontab` has mode `0o644`, owner `root`, group `root`. Permissions are compliant (`<= 0o640` or `0o600`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 2.4.1.3: Ensure access to /etc/cron.hourly is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.hourly` has mode `0o755`, owner `root`, group `root`. Permissions are compliant (`<= 0o755`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 2.4.1.4: Ensure access to /etc/cron.daily is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.daily` has mode `0o755`, owner `root`, group `root`. Permissions are compliant (`<= 0o755`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 2.4.1.5: Ensure access to /etc/cron.weekly is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.weekly` has mode `0o755`, owner `root`, group `root`. Permissions are compliant (`<= 0o755`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 2.4.1.6: Ensure access to /etc/cron.monthly is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.monthly` has mode `0o755`, owner `root`, group `root`. Permissions are compliant (`<= 0o755`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 2.4.1.7: Ensure access to `/etc/cron.yearly` is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.yearly` has mode `0o755`, owner `root`, group `root`. Permissions are compliant (`<= 0o755`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 2.4.1.8: Ensure access to `/etc/cron.d` is configured

- **Status:** PASS
- **Evidence:** `/etc/cron.d` has mode `0o755`, owner `root`, group `root`. Permissions are compliant (`<= 0o755`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 2.4.1.9: Ensure access to crontab mechanism is configured

- **Status:** PASS
- **Evidence:** `/var/spool/cron/crontabs` has mode `0o1730`, owner `root`, group `crontab`. `/usr/bin/crontab` has mode `0o2755`, owner `root`, group `crontab`. All permissions are compliant.
- **Recommendation:** None.

Rule 2.4.2.1: Ensure access to `at` is configured

- **Status:** NOT APPLICABLE
- **Evidence:** `at_installed` is `false`. The `at` package is not installed.
- **Recommendation:** None.

Section 3 – Network Configuration

Rule 3.1.1: Ensure IPv6 status is identified (Manual/Informational)

- **Status:** INFORMATIONAL
- **Evidence:**
 - IPv6 is enabled in `sysfs` (`sysfs_disable_value: "0"`).
 - IPv6 is not disabled in `GRUB` (`grub_ipv6_disable_flag: false`).
 - 2 IPv6 addresses are found on network interfaces (`inet6_address_count: 2`).
- **Recommendation:** Review if IPv6 is required for the system's function. If not, consider disabling it at the kernel level.

Rule 3.1.2: Ensure wireless interfaces are not available

- **Status:** FAIL
- **Evidence:** `any_wireless_found` is `true`. The `cfg80211` kernel module is loaded.
- **Recommendation:** Disable or remove wireless interfaces if not required. Blacklist the

```
cfg80211 kernel module: echo "blacklist cfg80211" >> /etc/modprobe.d/  
wireless.conf and rmmod cfg80211.
```

Rule 3.1.3: Ensure Bluetooth services are not in use

- **Status:** PASS
- **Evidence:** bluez_installed is false. Bluetooth packages are not installed.
- **Recommendation:** None.

Rule 3.2.1: Ensure atm network module is not enabled

- **Status:** PASS
- **Evidence:** Module atm is not currently loaded (currently_loaded: false). No modprobe_config_lines found.
- **Recommendation:** None.

Rule 3.2.2: Ensure can network module is not enabled

- **Status:** PASS
- **Evidence:** Module can is not currently loaded (currently_loaded: false). No modprobe_config_lines found.
- **Recommendation:** None.

Rule 3.2.3: Ensure dccp network module is not enabled

- **Status:** PASS
- **Evidence:** Module dccp is not currently loaded (currently_loaded: false). No modprobe_config_lines found.
- **Recommendation:** None.

Rule 3.2.4: Ensure rds network module is not enabled

- **Status:** PASS
- **Evidence:** Module rds is not currently loaded (currently_loaded: false). No modprobe_config_lines found.
- **Recommendation:** None.

Rule 3.2.5: Ensure sctp network module is not enabled

- **Status:** PASS
- **Evidence:** Module sctp is not currently loaded (currently_loaded: false). No modprobe_config_lines found.
- **Recommendation:** None.

Rule 3.2.6: Ensure tipc network module is not enabled

- **Status:** PASS
- **Evidence:** Module tipc is not currently loaded (currently_loaded: false). No modprobe_config_lines found.
- **Recommendation:** None.

Rule 3.3.1.1: Ensure net.ipv4.ip_forward is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.ip_forward runtime_value` is "0". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.1.2: Ensure net.ipv4.conf.all.forwarding is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.conf.all.forwarding runtime_value` is "0". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.1.3: Ensure net.ipv4.conf.default.forwarding is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.conf.default.forwarding runtime_value` is "0". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.1.4: Ensure net.ipv4.conf.all.send_redirects is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.send_redirects runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.all.send_redirects` to 0. Add `net.ipv4.conf.all.send_redirects=0` to a sysctl configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.5: Ensure net.ipv4.conf.default.send_redirects is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.send_redirects runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.default.send_redirects` to 0. Add `net.ipv4.conf.default.send_redirects=0` to a sysctl configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.6: Ensure net.ipv4.icmp_ignore_bogus_error_responses is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.icmp_ignore_bogus_error_responses runtime_value` is "1". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.1.7: Ensure net.ipv4.icmp_echo_ignore_broadcasts is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.icmp_echo_ignore_broadcasts runtime_value` is "1". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.1.8: Ensure net.ipv4.conf.all.accept_redirects is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.accept_redirects runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.all.accept_redirects` to 0. Add `net.ipv4.conf.all.accept_redirects=0` to a sysctl configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.9: Ensure net.ipv4.conf.default.accept_redirects is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.accept_redirects runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.default.accept_redirects` to 0. Add `net.ipv4.conf.default.accept_redirects=0` to a sysctl configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.10: Ensure net.ipv4.conf.all.secure_redirects is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.secure_redirects runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.all.secure_redirects` to 0. Add `net.ipv4.conf.all.secure_redirects=0` to a sysctl configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.11: Ensure net.ipv4.conf.default.secure_redirects is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.secure_redirects runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.default.secure_redirects` to 0. Add `net.ipv4.conf.default.secure_redirects=0` to a sysctl configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.12: Ensure net.ipv4.conf.all.rp_filter is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.conf.all.rp_filter runtime_value` is "2" and `persisted_value` is "2". Both "1" and "2" are compliant.
- **Recommendation:** None.

Rule 3.3.1.13: Ensure net.ipv4.conf.default.rp_filter is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.conf.default.rp_filter runtime_value` is "2" and `persisted_value` is "2". Both "1" and "2" are compliant.
- **Recommendation:** None.

Rule 3.3.1.14: Ensure net.ipv4.conf.all.accept_source_route is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.conf.all.accept_source_route runtime_value` is "0". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.1.15: Ensure net.ipv4.conf.default.accept_source_route is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.accept_source_route runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.default.accept_source_route` to 0. Add `net.ipv4.conf.default.accept_source_route=0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.16: Ensure net.ipv4.conf.all.log_martians is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.log_martians runtime_value` is "0". Expected value is "1". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.all.log_martians` to 1. Add `net.ipv4.conf.all.log_martians=1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.17: Ensure net.ipv4.conf.default.log_martians is configured

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.log_martians runtime_value` is "0". Expected value is "1". `persisted_value` is null.
- **Recommendation:** Set `net.ipv4.conf.default.log_martians` to 1. Add `net.ipv4.conf.default.log_martians=1` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.1.18: Ensure net.ipv4.tcp_syncookies is configured

- **Status:** PASS
- **Evidence:** `net.ipv4.tcp_syncookies runtime_value` is "1". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.2.1: Ensure net.ipv6.conf.all.forwarding is configured

- **Status:** PASS
- **Evidence:** `net.ipv6.conf.all.forwarding runtime_value` is "0". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.2.2: Ensure net.ipv6.conf.default.forwarding is configured

- **Status:** PASS
- **Evidence:** `net.ipv6.conf.default.forwarding runtime_value` is "0". `persisted_value` is null.

- **Recommendation:** None.

Rule 3.3.2.3: Ensure net.ipv6.conf.all.accept_redirects is configured

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.all.accept_redirects runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv6.conf.all.accept_redirects` to 0. Add `net.ipv6.conf.all.accept_redirects=0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.2.4: Ensure net.ipv6.conf.default.accept_redirects is configured

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.default.accept_redirects runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv6.conf.default.accept_redirects` to 0. Add `net.ipv6.conf.default.accept_redirects=0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.2.5: Ensure net.ipv6.conf.all.accept_source_route is configured

- **Status:** PASS
- **Evidence:** `net.ipv6.conf.all.accept_source_route runtime_value` is "0". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.2.6: Ensure net.ipv6.conf.default.accept_source_route is configured

- **Status:** PASS
- **Evidence:** `net.ipv6.conf.default.accept_source_route runtime_value` is "0". `persisted_value` is null.
- **Recommendation:** None.

Rule 3.3.2.7: Ensure net.ipv6.conf.all.accept_ra is configured

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.all.accept_ra runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv6.conf.all.accept_ra` to 0. Add `net.ipv6.conf.all.accept_ra=0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Rule 3.3.2.8: Ensure net.ipv6.conf.default.accept_ra is configured

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.default.accept_ra runtime_value` is "1". Expected value is "0". `persisted_value` is null.
- **Recommendation:** Set `net.ipv6.conf.default.accept_ra` to 0. Add `net.ipv6.conf.default.accept_ra=0` to a `sysctl` configuration file (e.g., `/etc/sysctl.d/99-sysctl.conf`) and apply with `sudo sysctl -p`.

Section 4.1 – Firewall

Rule 4.1.1: Ensure ufw is installed (Automated)

- **Status:** PASS
- **Evidence:** `ufw_installed` is `true` and `ufw_dpkg_status` shows "install ok installed".
- **Recommendation:** None.

Rule 4.1.2: Ensure ufw service is configured (Automated)

- **Status:** FAIL
- **Evidence:** `ufw` service is enabled but inactive (`status_verbos`: "Status: inactive"). No alternative firewalls are active.
- **Recommendation:** Enable and start the UFW service: `sudo ufw enable`.

Rule 4.1.3: Ensure ufw incoming default is configured (Automated)

- **Status:** PASS
- **Evidence:** `DEFAULT_INPUT_POLICY` is "DROP".
- **Recommendation:** None.

Rule 4.1.4: Ensure ufw outgoing default is configured (Automated)

- **Status:** FAIL
- **Evidence:** `DEFAULT_OUTPUT_POLICY` is "ACCEPT". CIS recommends "DENY" or "REJECT".
- **Recommendation:** Set the default outgoing policy for UFW to DENY or REJECT if not explicitly required to allow all outbound connections. This can be done by editing `/etc/default/ufw` or using `sudo ufw default deny outgoing`.

Rule 4.1.5: Ensure ufw routed default is configured (Automated)

- **Status:** PASS
- **Evidence:** `DEFAULT_FORWARD_POLICY` is "DROP".
- **Recommendation:** None.

Section 5.1 – Configure SSH Server

Rule 5.1.1: Ensure access to `/etc/ssh/sshd_config` is configured (Automated)

- **Status:** FAIL
- **Evidence:** `/etc/ssh/sshd_config` has mode `0o644`, owner `root`, group `root`. The recommended mode is `0o600`.
- **Recommendation:** Set permissions of `/etc/ssh/sshd_config` to `0o600`: `sudo chmod 0600 /etc/ssh/sshd_config`.

Rule 5.1.2: Ensure access to SSH private host key files is configured (Automated)

- **Status:** PASS
- **Evidence:**
 - `/etc/ssh/ssh_host_ecdsa_key` has mode `0o600`, owner `root`, group `root`.
 - `/etc/ssh/ssh_host_ed25519_key` has mode `0o600`, owner `root`, group `root`.
 - `/etc/ssh/ssh_host_rsa_key` has mode `0o600`, owner `root`, group `root`. All private key files have compliant permissions (`<= 0o600`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 5.1.3: Ensure access to SSH public host key files is configured (Automated)

- **Status:** PASS
- **Evidence:**
 - `/etc/ssh/ssh_host_ecdsa_key.pub` has mode `0o644`, owner `root`, group `root`.
 - `/etc/ssh/ssh_host_ed25519_key.pub` has mode `0o644`, owner `root`, group `root`.
 - `/etc/ssh/ssh_host_rsa_key.pub` has mode `0o644`, owner `root`, group `root`. All public key files have compliant permissions (`<= 0o644`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 5.1.4: Ensure sshd access is configured (Automated)

- **Status:** FAIL
- **Evidence:** `password_authentication` is `true` and `authentication_methods` is `"any"`. This allows password-based authentication, which is generally less secure than key-based authentication.
- **Recommendation:** Disable password authentication by setting `PasswordAuthentication no` in `/etc/ssh/sshd_config`. Consider restricting `AuthenticationMethods` to `publickey` only.

Rule 5.1.5: Ensure sshd Banner is configured (Automated)

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is `"none"`.
- **Recommendation:** Configure a warning banner for SSH by setting `Banner /etc/issue.net` (or a custom banner file) in `/etc/ssh/sshd_config`.

Rule 5.1.6: Ensure sshd Ciphers are configured (Automated)

- **Status:** PASS
- **Evidence:** The following ciphers are configured: `chacha20-poly1305@openssh.com`, `aes128-ctr`, `aes192-ctr`, `aes256-ctr`, `aes128-gcm@openssh.com`, `aes256-gcm@openssh.com`. These are generally considered strong ciphers.
- **Recommendation:** None.

Rule 5.1.7: Ensure sshd ClientAliveInterval and ClientAliveCountMax are configured (Automated)

- **Status:** FAIL
- **Evidence:** ClientAliveInterval is "0", which means no keep-alive messages are sent. ClientAliveCountMax is "3". CIS recommends ClientAliveInterval to be between 1 and 300 seconds.
- **Recommendation:** Set ClientAliveInterval to a value between 1 and 300 (e.g., 300) in /etc/ssh/sshd_config to ensure idle SSH sessions are terminated.

Rule 5.1.8: Ensure sshd DisableForwarding is enabled (Automated)

- **Status:** FAIL
- **Evidence:** AllowTcpForwarding is true, AllowAgentForwarding is true, and AllowStreamLocalForwarding is true. CIS recommends disabling all forwarding.
- **Recommendation:** Disable all forwarding by setting AllowTcpForwarding no, AllowAgentForwarding no, and AllowStreamLocalForwarding no in /etc/ssh/sshd_config.

Rule 5.1.9: Ensure sshd GSSAPIAuthentication is disabled (Automated)

- **Status:** PASS
- **Evidence:** GSSAPIAuthentication is false.
- **Recommendation:** None.

Rule 5.1.10: Ensure sshd HostbasedAuthentication is disabled (Automated)

- **Status:** PASS
- **Evidence:** HostbasedAuthentication is false.
- **Recommendation:** None.

Rule 5.1.11: Ensure sshd IgnoreRhosts is enabled (Automated)

- **Status:** PASS
- **Evidence:** IgnoreRhosts is true.
- **Recommendation:** None.

Rule 5.1.12: Ensure sshd KexAlgorithms is configured (Automated)

- **Status:** FAIL
- **Evidence:** The crypto.weak_kex list contains several weak key exchange algorithms: ecdh-sha2-nistp256, ecdh-sha2-nistp384, ecdh-sha2-nistp521 (due to NIST/NSA backdoor concerns) and diffie-hellman-group14-sha256 (deprecated/weaknesses).
- **Recommendation:** Remove weak KexAlgorithms from the KexAlgorithms directive in /etc/ssh/sshd_config. Consult CIS benchmark for a list of recommended KexAlgorithms.

Rule 5.1.13: Ensure sshd LoginGraceTime is configured (Automated)

- **Status:** PASS
- **Evidence:** `LoginGraceTime` is "120". This is within the recommended range (≥ 60 and ≤ 300).
- **Recommendation:** None.

Rule 5.1.14: Ensure sshd LogLevel is configured (Automated)

- **Status:** PASS
- **Evidence:** `LogLevel` is "INFO". This is compliant (INFO or VERBOSE).
- **Recommendation:** None.

Rule 5.1.15: Ensure sshd MACs are configured (Automated)

- **Status:** FAIL
- **Evidence:** The `crypto.weak_macs` list contains several weak MAC algorithms: `umac-64-etm@openssh.com`, `hmac-sha1-etm@openssh.com`, `umac-64@openssh.com`, `umac-128@openssh.com`, `hmac-sha2-256`, `hmac-sha2-512`, `hmac-sha1`.
- **Recommendation:** Remove weak MAC algorithms from the MACs directive in `/etc/ssh/sshd_config`. Consult CIS benchmark for a list of recommended MACs.

Rule 5.1.16: Ensure sshd MaxAuthTries is configured (Automated)

- **Status:** FAIL
- **Evidence:** `MaxAuthTries` is "6". CIS recommends a maximum of 4.
- **Recommendation:** Set `MaxAuthTries` to 4 or less in `/etc/ssh/sshd_config`.

Rule 5.1.17: Ensure sshd MaxStartups is configured (Automated)

- **Status:** FAIL
- **Evidence:** `MaxStartups` is "10:30:100". CIS recommends "10:30:60".
- **Recommendation:** Set `MaxStartups` to 10:30:60 in `/etc/ssh/sshd_config`.

Rule 5.1.18: Ensure sshd MaxSessions is configured (Automated)

- **Status:** PASS
- **Evidence:** `MaxSessions` is "10". This is compliant (≤ 10).
- **Recommendation:** None.

Rule 5.1.19: Ensure sshd PermitEmptyPasswords is disabled (Automated)

- **Status:** PASS
- **Evidence:** `PermitEmptyPasswords` is false.
- **Recommendation:** None.

Rule 5.1.20: Ensure sshd PermitRootLogin is disabled (Automated)

- **Status:** FAIL
- **Evidence:** `PermitRootLogin` is "without-password". CIS recommends "no", "prohibit-password", or "forced-commands-only". "without-password" still allows root login with other methods (e.g., public key).
- **Recommendation:** Set `PermitRootLogin` no in `/etc/ssh/sshd_config` to

completely disable direct root login via SSH.

Rule 5.1.21: Ensure sshd PermitUserEnvironment is disabled (Automated)

- **Status:** PASS
- **Evidence:** `PermitUserEnvironment` is `false`.
- **Recommendation:** None.

Rule 5.1.22: Ensure sshd UsePAM is enabled (Automated)

- **Status:** PASS
- **Evidence:** `UsePAM` is `true`.
- **Recommendation:** None.

Rule 5.1.23: Ensure sshd post-quantum cryptography key exchange algorithms are configured (Automated)

- **Status:** UNKNOWN
- **Evidence:** The security profile does not provide specific information about the configuration of post-quantum cryptography key exchange algorithms.
- **Recommendation:** Manually verify if post-quantum cryptography key exchange algorithms are configured according to current best practices and CIS recommendations.

Rule 5.1.24: Ensure sshd ListenAddress is configured (Automated)

- **Status:** FAIL
- **Evidence:** `ListenAddress` is `"0.0.0.0:22"`. This listens on all available IPv4 interfaces.
- **Recommendation:** Configure `ListenAddress` to a specific IP address (e.g., `ListenAddress 192.168.1.10:22`) in `/etc/ssh/sshd_config` to restrict SSH access to a specific network interface.

Section 5.2 – Configure privilege escalation

Rule 5.2.1: Ensure sudo is installed (Automated)

- **Status:** PASS
- **Evidence:** `sudo_installed` is `true` with version `"1.9.15p5"`.
- **Recommendation:** None.

Rule 5.2.2: Ensure sudo commands use pty (Automated)

- **Status:** PASS
- **Evidence:** `use_pty_entries` contains `"Defaults use_pty"` from `/etc/sudoers`.
- **Recommendation:** None.

Rule 5.2.3: Ensure sudo log file exists (Automated)

- **Status:** PASS
- **Evidence:** `sudo_syslog_logging.auth_log_exists` is `true` and `syslog_log_exists` is `true`. Sudo logging is configured via `rsyslog`.

- **Recommendation:** None.

Rule 5.2.4: Ensure users must provide password for escalation (Automated)

- **Status:** PASS
- **Evidence:** `nopasswd_entries` is empty and `noauthenticate_entries` is empty. This indicates no global NOPASSWD or NOAUTHENTICATE directives are set.
- **Recommendation:** None.

Rule 5.2.5: Ensure re-authentication for privilege escalation is not disabled globally (Automated)

- **Status:** PASS
- **Evidence:** `noauthenticate_entries` is empty. This indicates re-authentication is not disabled globally.
- **Recommendation:** None.

Rule 5.2.6: Ensure sudo timestamp_timeout is configured (Automated)

- **Status:** FAIL
- **Evidence:** `timestamp_timeout_entries` is empty. No Defaults `timestamp_timeout` directive is configured.
- **Recommendation:** Add Defaults `timestamp_timeout=X` (where X is a positive integer, e.g., 15) to `/etc/sudoers` or a file in `/etc/sudoers.d/` to require re-authentication after a period of inactivity.

Rule 5.2.7: Ensure access to the su command is restricted (Automated)

- **Status:** FAIL
- **Evidence:** The `pam_wheel.so` line in `/etc/pam.d/su` is commented out (`# auth required pam_wheel.so`), meaning `su` access is not restricted to members of the `wheel` group.
- **Recommendation:** Uncomment the `auth required pam_wheel.so` line in `/etc/pam.d/su` to restrict `su` command usage to members of the `wheel` group. Ensure the `wheel` group exists and contains authorized users.

Section 5.3 – Pluggable Authentication Modules (PAM)

Rule 5.3.1.1: Ensure libpam-runtime is installed

- **Status:** PASS
- **Evidence:** `libpam-runtime` is installed (`true`) and `installed_version` ("1.5.3-5ubuntu5.5") matches `candidate_version` ("1.5.3-5ubuntu5.5").
- **Recommendation:** None.

Rule 5.3.1.2: Ensure libpam-modules is installed

- **Status:** PASS
- **Evidence:** `libpam-modules` is installed (`true`) and `installed_version` ("1.5.3-5ubuntu5.5") matches `candidate_version` ("1.5.3-5ubuntu5.5").

- **Recommendation:** None.

Rule 5.3.1.3: Ensure libpam-pwquality is installed

- **Status:** FAIL
- **Evidence:** libpam-pwquality is not installed (false).
- **Recommendation:** Install the libpam-pwquality package: `sudo apt install libpam-pwquality`.

Rule 5.3.1.4: Ensure cracklib-runtime is installed

- **Status:** FAIL
- **Evidence:** cracklib-runtime is not installed (false).
- **Recommendation:** Install the cracklib-runtime package: `sudo apt install cracklib-runtime`.

Rule 5.3.2.1: Ensure pam_unix is enabled

- **Status:** PASS
- **Evidence:** pam_unix.so is present in common-auth, common-account, common-password, and common-session PAM configuration files.
- **Recommendation:** None.

Rule 5.3.2.2: Ensure pam_faillock is enabled

- **Status:** FAIL
- **Evidence:** pam_faillock.so is not present in any PAM configuration files (pam_faillock.faillock_pam_lines is empty).
- **Recommendation:** Enable pam_faillock by adding appropriate lines to /etc/pam.d/common-auth and /etc/pam.d/common-account to lock accounts after failed login attempts.

Rule 5.3.2.3: Ensure pam_pwquality is enabled

- **Status:** FAIL
- **Evidence:** pam_pwquality.so is not present in any PAM configuration files (pam_pwquality.pwquality_pam_lines is empty).
- **Recommendation:** Enable pam_pwquality by adding password requisite pam_pwquality.so to /etc/pam.d/common-password to enforce strong password policies.

Rule 5.3.2.4: Ensure pam_pwhistory is enabled

- **Status:** FAIL
- **Evidence:** pam_pwhistory.so is not present in any PAM configuration files (pam_pwhistory.pwhistory_pam_lines is empty).
- **Recommendation:** Enable pam_pwhistory by adding password requisite pam_pwhistory.so to /etc/pam.d/common-password to enforce password history.

Rule 5.3.3.1.1: Ensure pam_faillock deny is configured

- **Status:** FAIL
- **Evidence:** pam_faillock is not enabled. The deny parameter is not configured.
- **Recommendation:** Enable pam_faillock and configure deny = 5 (or less) in /etc/security/faillock.conf or as an inline argument in /etc/pam.d/common-auth.

Rule 5.3.3.1.2: Ensure pam_faillock unlock_time is configured

- **Status:** FAIL
- **Evidence:** pam_faillock is not enabled. The unlock_time parameter is not configured.
- **Recommendation:** Enable pam_faillock and configure unlock_time = 900 (or 0 for never unlock) in /etc/security/faillock.conf or as an inline argument in /etc/pam.d/common-auth.

Rule 5.3.3.1.3: Ensure pam_faillock even_deny_root / root_unlock_time is configured

- **Status:** FAIL
- **Evidence:** pam_faillock is not enabled. even_deny_root and root_unlock_time parameters are not configured.
- **Recommendation:** Enable pam_faillock and configure even_deny_root and root_unlock_time = 60 (or greater) in /etc/security/faillock.conf or as inline arguments in /etc/pam.d/common-auth.

Rule 5.3.3.2.1: Ensure pam_pwquality difok is configured

- **Status:** FAIL
- **Evidence:** pam_pwquality is not enabled. The difok parameter is not configured.
- **Recommendation:** Enable pam_pwquality and configure difok = 2 (or greater) in /etc/security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.2.2: Ensure pam_pwquality minlen is configured

- **Status:** FAIL
- **Evidence:** pam_pwquality is not enabled. The minlen parameter is not configured.
- **Recommendation:** Enable pam_pwquality and configure minlen = 14 (or greater) in /etc/security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.2.3: Ensure pam_pwquality password complexity is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** pam_pwquality is not enabled, so dcredit, ucredit, lcredit, ocredit are not configured.
- **Recommendation:** Enable pam_pwquality and configure password complexity requirements (e.g., dcredit=-1, ucredit=-1, lcredit=-1, ocredit=-1 for at least one digit, uppercase, lowercase, and other character) in /etc/security/pwquality.conf or as inline arguments in /etc/pam.d/common-password.

Rule 5.3.3.2.4: Ensure pam_pwquality maxrepeat is configured

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled. The `maxrepeat` parameter is not configured.
- **Recommendation:** Enable `pam_pwquality` and configure `maxrepeat = 3` (or less) in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.5: Ensure pam_pwquality maxsequence is configured

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled. The `maxsequence` parameter is not configured.
- **Recommendation:** Enable `pam_pwquality` and configure `maxsequence = 3` (or less) in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.6: Ensure pam_pwquality dictcheck is configured

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled. The `dictcheck` parameter is not configured.
- **Recommendation:** Enable `pam_pwquality` and configure `dictcheck = 1` in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.7: Ensure pam_pwquality enforcing is configured

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled. The `enforcing` parameter is not configured.
- **Recommendation:** Enable `pam_pwquality` and configure `enforcing = 1` in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.2.8: Ensure pam_pwquality enforce_for_root is configured

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled. The `enforce_for_root` parameter is not configured.
- **Recommendation:** Enable `pam_pwquality` and configure `enforce_for_root` in `/etc/security/pwquality.conf` or as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.3.1: Ensure pam_pwhistory remember is configured

- **Status:** FAIL
- **Evidence:** `pam_pwhistory` is not enabled. The `remember` parameter is not configured.
- **Recommendation:** Enable `pam_pwhistory` and configure `remember = 24` (or greater) as an inline argument in `/etc/pam.d/common-password`.

Rule 5.3.3.3.2: Ensure pam_pwhistory enforce_for_root is configured

- **Status:** FAIL
- **Evidence:** pam_pwhistory is not enabled. The enforce_for_root parameter is not configured.
- **Recommendation:** Enable pam_pwhistory and configure enforce_for_root as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.3.3: Ensure pam_pwhistory use_authtok is configured

- **Status:** FAIL
- **Evidence:** pam_pwhistory is not enabled. The use_authtok parameter is not configured.
- **Recommendation:** Enable pam_pwhistory and configure use_authtok as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.4.1: Ensure pam_unix nullok is absent

- **Status:** FAIL
- **Evidence:** pam_unix.so in /etc/pam.d/common-auth includes the nullok flag.
- **Recommendation:** Remove the nullok flag from the pam_unix.so line in /etc/pam.d/common-auth to prevent users from logging in with empty passwords.

Rule 5.3.3.4.2: Ensure pam_unix remember is absent

- **Status:** PASS
- **Evidence:** The remember flag is not present in any pam_unix.so line.
- **Recommendation:** None.

Rule 5.3.3.4.3: Ensure pam_unix strong hashing algorithm is configured

- **Status:** PASS
- **Evidence:** pam_unix.so in /etc/pam.d/common-password includes the yescrypt flag.
- **Recommendation:** None.

Rule 5.3.3.4.4: Ensure pam_unix use_authtok is present (in password stack)

- **Status:** FAIL
- **Evidence:** The use_authtok flag is not present in any pam_unix.so line in the password stack.
- **Recommendation:** Add the use_authtok flag to the pam_unix.so line in /etc/pam.d/common-password to ensure that password changes are applied to all stacked modules.

Section 5.4 – User Accounts and Environment

Rule 5.4.1.1: Ensure password expiration is configured (Automated)

- **Status:** FAIL

- **Evidence:** PASS_MAX_DAYS is set to "99999" in /etc/login.defs. This effectively disables password expiration.
- **Recommendation:** Set PASS_MAX_DAYS to 365 or less in /etc/login.defs to enforce regular password changes.

Rule 5.4.1.2: Ensure minimum password days is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** PASS_MIN_DAYS is set to "0" in /etc/login.defs.
- **Recommendation:** Review and set PASS_MIN_DAYS to a value greater than 0 (e.g., 7) in /etc/login.defs to prevent immediate password changes.

Rule 5.4.1.3: Ensure password expiration warning days is configured (Automated)

- **Status:** PASS
- **Evidence:** PASS_WARN_AGE is set to "7" in /etc/login.defs. This is compliant (≥ 7).
- **Recommendation:** None.

Rule 5.4.1.4: Ensure strong password hashing algorithm is configured (Automated)

- **Status:** PASS
- **Evidence:** ENCRYPT_METHOD is set to "SHA512" in /etc/login.defs. pam_unix.so also uses yescrypt. Both are strong hashing algorithms.
- **Recommendation:** None.

Rule 5.4.1.5: Ensure inactive password lock is configured (Automated)

- **Status:** FAIL
- **Evidence:** useradd_inactive is set to "-1", which means accounts are never locked due to inactivity.
- **Recommendation:** Set INACTIVE to 30 (or a suitable value) in /etc/default/useradd and ensure useradd_inactive is configured to a positive value (e.g., 30) to lock inactive accounts.

Rule 5.4.1.6: Ensure all users last password change date is in the past (Automated)

- **Status:** FAIL
- **Evidence:** Several users (e.g., root, daemon, bin, sys, sync, games, man, lp, mail, news, uucp, proxy, www-data, backup, list, irc, _apt, nobody, systemd-network, systemd-timesync, dhcpcd, messagebus, systemd-resolve, pollinate, polkitd, syslog, uidd, tcpdump, tss, landscape, fwupd-refresh) have their last_password_change date set to a future date (Feb 16, 2025, Apr 19, 2026, Jun 25, 2026).
- **Recommendation:** Force a password change for all users with future password change dates using `sudo chage -d 0 <username>` or ensure that the system's clock is correctly synchronized and that password changes are properly recorded.

Rule 5.4.2.1: Ensure root is the only UID 0 account (Automated)

- **Status:** PASS
- **Evidence:** Only "root" is listed as a UID 0 account.
- **Recommendation:** None.

Rule 5.4.2.2: Ensure root is the only GID 0 account (Automated)

- **Status:** PASS
- **Evidence:** Only "root" is listed as a primary GID 0 account.
- **Recommendation:** None.

Rule 5.4.2.3: Ensure group root is the only GID 0 group (Automated)

- **Status:** PASS
- **Evidence:** Only "root" is listed as a GID 0 group.
- **Recommendation:** None.

Rule 5.4.2.4: Ensure root account access is controlled (Automated)

- **Status:** FAIL
- **Evidence:** `root_shadow_password_status` is "*", indicating the root account is locked. However, `securetty_exists` is false, meaning `/etc/securetty` does not exist to restrict root logins to specific TTYs.
- **Recommendation:** Create `/etc/securetty` and list only the console TTYs from which root is allowed to log in directly.

Rule 5.4.2.5: Ensure root path integrity (Automated)

- **Status:** UNKNOWN
- **Evidence:** The root path is `/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin`. The profile does not provide sufficient information to verify the permissions of all directories in the root's PATH.
- **Recommendation:** Manually verify that all directories in the root user's PATH are owned by root and are not world-writable or group-writable.

Rule 5.4.2.6: Ensure root user umask is configured (Automated)

- **Status:** FAIL
- **Evidence:** The root umask is "022" from `/etc/login.defs`. CIS recommends a umask of "027" or "077".
- **Recommendation:** Set the root user's umask to 027 or 077 in `/etc/login.defs` and in `/root/.bashrc` or `/root/.profile`.

Rule 5.4.2.7: Ensure system accounts do not have a valid login shell (Automated)

- **Status:** FAIL
- **Evidence:** The `sync` account has `/bin/sync` as its shell, which is a valid login shell.
- **Recommendation:** Change the shell for the `sync` account to `/usr/sbin/nologin` or `/bin/false` using `sudo usermod -s /usr/sbin/nologin sync`.

Rule 5.4.2.8: Ensure accounts without a valid login shell are locked (Automated)

- **Status:** PASS
- **Evidence:** All system accounts listed with `/usr/sbin/nologin` or `/bin/false` as their shell have a `passwd_status` of "L" (locked).
- **Recommendation:** None.

Rule 5.4.3.1: Ensure nologin is not listed in /etc/shells (Automated)

- **Status:** PASS
- **Evidence:** `nologin_in_shells` is false. `/usr/sbin/nologin` is not listed in `/etc/shells`.
- **Recommendation:** None.

Rule 5.4.3.2: Ensure default user shell timeout is configured (Automated)

- **Status:** FAIL
- **Evidence:** `tmout_settings` is empty for `/etc/profile`, `/etc/profile.d`, and `/etc/bash.bashrc`. No `TMOUT` setting was found.
- **Recommendation:** Add `TMOUT=600` (or a suitable value) to `/etc/profile.d/timeout.sh` to automatically log out inactive shell sessions.

Rule 5.4.3.3: Ensure default user umask is configured (Automated)

- **Status:** FAIL
- **Evidence:** The default user umask is "022" from `/etc/login.defs`. CIS recommends a umask of "027" or "077".
- **Recommendation:** Set the default user umask to 027 or 077 in `/etc/login.defs` and in `/etc/bash.bashrc` or `/etc/profile`.

Section 6.1 – System Logging

Rule 6.1.1.1.1: Ensure systemd-journald service is enabled and active (Automated)

- **Status:** PASS
- **Evidence:** `systemd-journald.service` is static (enabled by default) and active.
- **Recommendation:** None.

Rule 6.1.1.1.2: Ensure systemd-journal-remote service is not in use (Automated)

- **Status:** PASS
- **Evidence:** `systemd-journal-remote` is not installed (false).
- **Recommendation:** None.

Rule 6.1.1.1.3: Ensure journald is configured to send logs to rsyslog (Automated)

- **Status:** FAIL
- **Evidence:** The `journald.conf` file does not explicitly set `ForwardToSyslog=yes`. The default is no.
- **Recommendation:** Set `ForwardToSyslog=yes` in `/etc/systemd/journald.conf` to ensure logs are forwarded to rsyslog.

Rule 6.1.1.1.4: Ensure journald log file access is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** Journal log directories `/run/log/journal` and `/var/log/journal` have mode 2755, owner root, group 999 (likely `systemd-journal`). Individual journal files have mode 0640, owner root, group 999.
- **Recommendation:** Review the permissions of journal log files and directories to ensure they are restricted to authorized users.

Rule 6.1.1.1.5: Ensure journald log file rotation is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The `journald.conf` file does not explicitly configure log rotation parameters like `SystemMaxUse`, `SystemMaxFileSize`, `MaxRetentionSec`, or `MaxFileSec`.
- **Recommendation:** Configure journald log rotation parameters in `/etc/systemd/journald.conf` to manage log file sizes and retention.

Rule 6.1.1.1.6: Ensure journald Storage is configured (Automated)

- **Status:** FAIL
- **Evidence:** The `journald.conf` file has `Storage=auto` commented out. The default `auto` stores logs persistently if `/var/log/journal` exists, which it does. However, the rule requires it to be explicitly configured.
- **Recommendation:** Uncomment and set `Storage=persistent` in `/etc/systemd/journald.conf` to explicitly ensure persistent storage of journal logs.

Rule 6.1.1.1.7: Ensure journald Compress is configured (Automated)

- **Status:** FAIL
- **Evidence:** The `journald.conf` file has `Compress=yes` commented out. The default is yes, but the rule requires it to be explicitly configured.
- **Recommendation:** Uncomment and set `Compress=yes` in `/etc/systemd/journald.conf` to explicitly enable compression for journal logs.

Rule 6.1.2.1: Ensure rsyslog is installed (Automated)

- **Status:** PASS
- **Evidence:** `rsyslog_installed` is true.
- **Recommendation:** None.

Rule 6.1.2.2: Ensure rsyslog service is enabled and active (Automated)

- **Status:** PASS
- **Evidence:** `rsyslog.service` is enabled and active.
- **Recommendation:** None.

Rule 6.1.2.3: Ensure rsyslog default file permissions are configured (Automated)

- **Status:** PASS
- **Evidence:** `/etc/rsyslog.conf` contains `$FileOwner syslog, $FileGroup adm,` and `$FileCreateMode 0640.`
- **Recommendation:** None.

Rule 6.1.2.4: Ensure rsyslog logging is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** Rsyslog is configured to log `auth, authpriv.*` to `/var/log/auth.log`, `*.*;auth,authpriv.none` to `/var/log/syslog`, `kern.*` to `/var/log/kern.log`, `mail.*` to `/var/log/mail.log`, and `mail.err` to `/var/log/mail.err`.
- **Recommendation:** Review rsyslog configuration files (`/etc/rsyslog.conf` and `/etc/rsyslog.d/*.conf`) to ensure all necessary system events are being logged and retained appropriately.

Rule 6.1.2.5: Ensure rsyslog is configured to send logs to a remote log host (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The provided rsyslog configuration files do not show explicit directives for sending logs to a remote log host.
- **Recommendation:** If remote logging is required, configure rsyslog to send logs to a centralized log management system.

Rule 6.1.2.6: Ensure rsyslog is not configured to receive logs from a remote client (Automated)

- **Status:** PASS
- **Evidence:** The `imudp` and `imtcp` modules are commented out in `/etc/rsyslog.conf`, indicating rsyslog is not configured to receive logs from remote clients.
- **Recommendation:** None.

Rule 6.1.2.7: Ensure logrotate is configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** `logrotate` is installed and configured via `/etc/logrotate.conf` and files in `/etc/logrotate.d/`. The `logrotate.timer` is enabled and active.
- **Recommendation:** Review logrotate configurations to ensure all critical log files are rotated, compressed, and retained according to policy.

Rule 6.1.2.8: Ensure rsyslog-gnutls is installed (Automated)

- **Status:** FAIL
- **Evidence:** `rsyslog-gnutls` is not installed (`false`).
- **Recommendation:** Install the `rsyslog-gnutls` package: `sudo apt install rsyslog-gnutls.`

Rule 6.1.2.9: Ensure rsyslog is configured to use gtls for forwarding (Automated)

- **Status:** INFORMATIONAL
- **Evidence:** `rsyslog-gnutls` is not installed, so GTLS for forwarding is not configured.
- **Recommendation:** Install `rsyslog-gnutls` and configure `rsyslog` to use GTLS for secure remote log forwarding.

Rule 6.1.2.10: Ensure rsyslog CA certificates are configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** `rsyslog-gnutls` is not installed, so CA certificates for GTLS are not configured.
- **Recommendation:** If remote logging with GTLS is used, configure and manage CA certificates for secure communication.

Rule 6.1.3.1: Ensure access to all logfiles has been configured (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** A list of log files and their permissions is provided. Examples include `/var/log/auth.log` (0640 syslog adm), `/var/log/syslog` (0640 syslog adm), `/var/log/wtmp` (0664 root utmp), `/var/log/btmp` (0660 root utmp).
- **Recommendation:** Review the permissions of all log files to ensure they are restricted to authorized users and groups, preventing unauthorized access or modification.

Section 6.2 – System Auditing

Rule 6.2.1.1.1: Ensure auditd is installed (Automated)

- **Status:** FAIL
- **Evidence:** `auditd_installed` is false. The `auditd` package is not installed.
- **Recommendation:** Install the `auditd` package: `sudo apt install auditd audispd-plugins`.

Rule 6.2.1.1.2: Ensure auditd service is enabled and active (Automated)

- **Status:** FAIL
- **Evidence:** `auditd_installed` is false. The `auditd` service is not enabled or active.
- **Recommendation:** Install `auditd` and ensure its service is enabled and active: `sudo systemctl enable auditd && sudo systemctl start auditd`.

Rule 6.2.1.1.3: Ensure auditing for processes that start prior to auditd is enabled (Automated)

- **Status:** INFORMATIONAL
- **Evidence:** `auditd` is not installed. This check typically involves verifying `audit=1` in GRUB.
- **Recommendation:** Install `auditd`. Then, ensure `audit=1` is added to the kernel command line in `/etc/default/grub`, followed by `sudo update-grub` and a reboot.

Rule 6.2.1.1.4: Ensure audit_backlog_limit is sufficient (Automated)

- **Status:** INFORMATIONAL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and configure audit_backlog_limit in /etc/audit/auditd.conf to a sufficient value (e.g., 8192 or higher) to prevent event loss during high activity.

Rule 6.2.1.2.1: Ensure audit log storage size is configured (Automated)

- **Status:** INFORMATIONAL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and configure max_log_file in /etc/audit/auditd.conf to manage the size of audit log files.

Rule 6.2.1.2.2: Ensure audit logs are not automatically deleted (Automated)

- **Status:** INFORMATIONAL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and configure max_log_file_action = keep_logs in /etc/audit/auditd.conf to prevent automatic deletion of audit logs.

Rule 6.2.1.2.3: Ensure system is disabled when audit logs are full (Automated)

- **Status:** INFORMATIONAL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and configure disk_full_action = halt (or single) and disk_error_action = halt (or single) in /etc/audit/auditd.conf to ensure the system reacts appropriately when audit logs are full.

Rule 6.2.1.2.4: Ensure system warns when audit logs are low on space (Automated)

- **Status:** INFORMATIONAL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and configure space_left_action = email (or syslog) and space_left = 75% (or a suitable value) in /etc/audit/auditd.conf to warn when audit logs are low on space.

Rule 6.2.3.1-6.2.3.30: Ensure auditd rules cover sudoers, identity, logins, session, time-change, MAC policy, access, mounts, deletion, loading modules, and configuration is immutable (Automated)

- **Status:** FAIL
- **Evidence:** auditd_installed is false. Audit rules cannot be enforced or verified without auditd installed.
- **Recommendation:** Install auditd and configure a comprehensive set of audit rules as specified in the CIS Benchmark to cover critical system activities.

Rule 6.2.4.1-6.2.4.10: Ensure audit log files and audit tools file access and permissions are strictly constrained (Automated)

- **Status:** FAIL
- **Evidence:** `auditd_installed` is false. Audit log files and tools cannot be properly secured without `auditd` installed.
- **Recommendation:** Install `auditd` and ensure that audit log files (e.g., `/var/log/audit/audit.log`) and audit tools (`/sbin/auditctl`, `/sbin/audispd`, etc.) have strict permissions (e.g., `0o640` for logs, `0o755` for tools, owned by `root:root` or `root:audit`).

Section 6.3 – Integrity Checking

Rule 6.3.1: Ensure AIDE is installed (Automated)

- **Status:** FAIL
- **Evidence:** `aide_installed` is false. The AIDE package is not installed.
- **Recommendation:** Install the AIDE package: `sudo apt install aide aide-common`.

Rule 6.3.2: Ensure filesystem integrity is regularly checked (Automated)

- **Status:** FAIL
- **Evidence:** `aide_installed` is false. No scheduled integrity checking is configured (`scheduled_checking: null`).
- **Recommendation:** Install AIDE and configure a cron job or systemd timer to regularly run AIDE checks (e.g., daily or weekly).

Rule 6.3.3: Ensure cryptographic mechanisms protect tools (Automated)

- **Status:** FAIL
- **Evidence:** `aide_installed` is false. No integrity tracking for audit tools is configured (`audit_tools_integrity_tracked: null`).
- **Recommendation:** Install AIDE and configure its database to include critical system binaries and audit tools, ensuring they are protected with strong cryptographic hashes (e.g., SHA512).

Section 7.1 – File Permissions

Rule 7.1.1: Ensure access to `/etc/passwd` is configured (Automated)

- **Status:** PASS
- **Evidence:** `/etc/passwd` has mode `0o644`, owner `root`, group `root`. Permissions are compliant (`<= 0o644`, owner `root`, group `root`).
- **Recommendation:** None.

Rule 7.1.2: Ensure access to `/etc/passwd-` is configured (Automated)

- **Status:** PASS
- **Evidence:** `/etc/passwd-` has mode `0o644`, owner `root`, group `root`. Permissions are compliant (`<= 0o644`, owner `root`, group `root`).

- **Recommendation:** None.

Rule 7.1.3: Ensure access to /etc/group is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/group has mode 0o644, owner root, group root. Permissions are compliant (< = 0o644, owner root, group root).
- **Recommendation:** None.

Rule 7.1.4: Ensure access to /etc/group- is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/group- has mode 0o644, owner root, group root. Permissions are compliant (< = 0o644, owner root, group root).
- **Recommendation:** None.

Rule 7.1.5: Ensure access to /etc/shadow is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/shadow has mode 0o640, owner root, group shadow. Permissions are compliant (< = 0o640, owner root, group shadow).
- **Recommendation:** None.

Rule 7.1.6: Ensure access to /etc/shadow- is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/shadow- has mode 0o640, owner root, group shadow. Permissions are compliant (< = 0o640, owner root, group shadow).
- **Recommendation:** None.

Rule 7.1.7: Ensure access to /etc/gshadow is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/gshadow has mode 0o640, owner root, group shadow. Permissions are compliant (< = 0o640, owner root, group shadow).
- **Recommendation:** None.

Rule 7.1.8: Ensure access to /etc/gshadow- is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/gshadow- has mode 0o640, owner root, group shadow. Permissions are compliant (< = 0o640, owner root, group shadow).
- **Recommendation:** None.

Rule 7.1.9: Ensure access to /etc/shells is configured (Automated)

- **Status:** PASS
- **Evidence:** /etc/shells has mode 0o644, owner root, group root. Permissions are compliant (< = 0o644, owner root, group root).
- **Recommendation:** None.

Rule 7.1.10: Ensure access to /etc/security/opasswd is configured

(Automated)

- **Status:** PASS
- **Evidence:** /etc/security/opasswd has mode 0o600, owner root, group root. Permissions are compliant ($\leq 0o600$, owner root, group root).
- **Recommendation:** None.

Rule 7.1.11: Ensure world writable files and directories are secured (Automated)

- **Status:** PASS
- **Evidence:** world_writable_files is empty and world_writable_dirs_no_sticky is empty. No world-writable files or directories without the sticky bit were found in scanned mounts.
- **Recommendation:** None.

Rule 7.1.12: Ensure no files or directories without an owner and a group exist (Automated)

- **Status:** PASS
- **Evidence:** unowned_or_ungrouped is empty. No files or directories without an owner or group were found in scanned mounts.
- **Recommendation:** None.

Rule 7.1.13: Ensure SUID and SGID files are reviewed (Manual)

- **Status:** INFORMATIONAL
- **Evidence:** The following SUID/SGID files were found:
 - /usr/bin/umount (4755 root root)
 - /usr/bin/chsh (4755 root root)
 - /usr/bin/sudo (4755 root root)
 - /usr/bin/su (4755 root root)
 - /usr/bin/passwd (4755 root root)
 - /usr/bin/ssh-agent (2755 root _ssh)
 - /usr/bin/fusermount3 (4755 root root)
 - /usr/bin/newgrp (4755 root root)
 - /usr/bin/chage (2755 root shadow)
 - /usr/bin/expiry (2755 root shadow)
 - /usr/bin/gpasswd (4755 root root)
 - /usr/bin/crontab (2755 root crontab)
 - /usr/bin/mount (4755 root root)
 - /usr/bin/chfn (4755 root root)
 - /usr/lib/dbus-1.0/dbus-daemon-launch-helper (4754 root messagebus)
 - /usr/lib/x86_64-linux-gnu/utempter/utempter (2755 root utmp)
 - /usr/lib/openssh/ssh-keysign (4755 root root)
 - /usr/lib/polkit-1/polkit-agent-helper-1 (4755 root root)
 - /usr/sbin/pam_extrausers_chkpwd (2755 root shadow)
 - /usr/sbin/unix_chkpwd (2755 root shadow)
- **Recommendation:** Regularly review all SUID/SGID files to ensure they are legitimate

and necessary for system operation. Remove SUID/SGID permissions from any files that do not require them.

Section 7.2 – Local User and Group Settings

Rule 7.2.1: Ensure accounts in `/etc/passwd` use shadowed passwords (Automated)

- **Status:** PASS
- **Evidence:** All user accounts listed in `user_chage_info` have either a locked status (L or *) or a valid password change date, indicating they use shadowed passwords.
- **Recommendation:** None.

Rule 7.2.2: Ensure `/etc/shadow` password fields are not empty (Automated)

- **Status:** PASS
- **Evidence:** The `root_shadow_password_status` is "*", which means the account is locked, not that the password field is empty. All other user entries in `user_chage_info` show a last password change date, implying non-empty password fields.
- **Recommendation:** None.

Rule 7.2.3: Ensure all groups in `/etc/passwd` exist in `/etc/group` (Automated)

- **Status:** PASS
- **Evidence:** `orphan_gids_in_passwd` is empty. All groups referenced in `/etc/passwd` exist in `/etc/group`.
- **Recommendation:** None.

Rule 7.2.4: Ensure shadow group is empty (Automated)

- **Status:** PASS
- **Evidence:** `shadow_group_members` is empty. The shadow group contains no members.
- **Recommendation:** None.

Rule 7.2.5: Ensure no duplicate UIDs exist (Automated)

- **Status:** PASS
- **Evidence:** `duplicate_uids` is empty. No duplicate UIDs were found.
- **Recommendation:** None.

Rule 7.2.6: Ensure no duplicate GIDs exist (Automated)

- **Status:** PASS
- **Evidence:** `duplicate_gids` is empty. No duplicate GIDs were found.
- **Recommendation:** None.

Rule 7.2.7: Ensure no duplicate user names exist (Automated)

- **Status:** PASS
- **Evidence:** duplicate_usernames is empty. No duplicate user names were found.
- **Recommendation:** None.

Rule 7.2.8: Ensure no duplicate group names exist (Automated)

- **Status:** PASS
- **Evidence:** duplicate_groupnames is empty. No duplicate group names were found.
- **Recommendation:** None.

Rule 7.2.9: Ensure local interactive user home directories are configured (Automated)

- **Status:** PASS
- **Evidence:** home_directory_issues is empty. All local interactive users have properly configured home directories.
- **Recommendation:** None.

Rule 7.2.10: Ensure local interactive user dot files access is configured (Automated)

- **Status:** PASS
- **Evidence:** dotfile_permission_issues is empty. No issues with dot file permissions were found.
- **Recommendation:** None.

SUID/SGID Triage Report

```
{
  "findings": [],
  "summary": {
    "suspicious_found": 0,
    "message": "No unexpected SUID/SGID files detected."
  }
}
```